The Operations Data Flywheel: Streaming Weak-Form Laws for Industrial Robotics, without Backpropagation or Batch Re-fitting

Preprint — every quantitative claim is the measured output of the accompanying experiment suite (`robotics.json`); results are means over 3 seeds of a 2-DOF arm logged at 500 Hz.

Abstract

Industrial robots are surrounded by the data their own operations generate — joint angles, velocities, commanded torques — yet the physics embedded in that stream is thrown away. State-of-the-art identification re-fits a frozen model in batch, offline, on curated data; a change in payload, wear, or contact invalidates the model until the next expensive re-fit. Here we show that the streaming weak-form operator from companion work—in which governing laws are identified by integrating them against a one-pole IIR test function, transferring derivatives off the noisy data by integration by parts—closes the loop: a robot can identify its own rigid-body law online, from the operations stream itself, and feed it back as feedforward computation in a closed flywheel. Measured on a 2-DOF planar arm with gravity, Coulomb friction, variable payload, and a socket-insertion contact task: at 5% observational noise the weak-form identifier recovers the law direction to **3.4**° (finite differences: **9.8**°, batch oracle: **4.3**°) and predicts commanded torque **3.7**× more accurately than finite differences. Closing the loop, a five-lap flywheel drops tracking error **7.1**× (NMSE 0.61→0.086) within two laps, while the same loop driven by finite differences stalls; when a 1.5 kg payload is added mid-experiment, the weak-form law re-adapts within one lap (FD's law direction degrades to **50.8**°, weak-form: **24**°). Across a six-robot fleet the identified law vectors separate payload and wear variants (direction cosine 0.99) and read payload mass to within 17%, and the frozen nominal law doubles as a contact detector with **31**× residual SNR. The loop needs no labels, no batches, and no backpropagation: it is the engine a robotics data flywheel is built on.

1 Introduction

1.1 The flywheel bottleneck

The central promise of the robotics data flywheel is that operational data accumulate into better models, and better models into better operation. Mind Robotics—Rivian's spinout for industrial AI—states the thesis explicitly: leverage manufacturing and operations data as the foundation of a robotics flywheel[1]. The premise is sound: a robot arm executing a weld, a pick, or an insertion produces a continuous stream of $\{q, \dot{q}, \tau\}$ at 500–1000 Hz, and that stream is a physical record of the exact plant running that exact task.

The flywheel, however, is blocked on the identification step. Standard practice re-fits dynamics models offline in batch[2, 3]: log data, stop, run least squares, deploy. The loop does not spin—it ratchets, because any change of payload, tooling, friction (wear), or contact invalidates the deployed model, and re-fitting is expensive and rare. Online alternatives that differentiate the stream (finite differences) amplify sensor noise by a factor $\sim 2/\Delta t^2$; at a 2 ms control period that is a 5×10^5 noise gain, which is why online identification in industry is usually restricted to slow, curated excitation trajectories rather than the operations stream itself.

1.2 The weak form as the online operator

The companion paper[7] introduced a strictly local, streaming, differentiation-free identification operator: every governing equation is integrated against a one-pole IIR test function ψ (window operator W), so time derivatives are transferred from the noisy data onto the analytic filter by integration by parts. Where finite differences divide by Δt^2, the weak form divides by λ^2 — the window decay rate — reducing noise gain by orders of magnitude while keeping the loop causal, recursive, and $O(1)$ in memory per quantity.

This paper transfers that operator from biomorphic networks to rigid-body robotics, where the law to be identified is the classical momentum form

$$\tau = \frac{d}{dt}\left[M(q)\,\dot{q}\right] + g(q) + f_{\text{fric}}(\dot{q}), \tag{1}$$

with M the inertia matrix, g gravity, and f_{fric} viscous plus Coulomb friction. Applying the window operator W to both sides — which is legal because W is linear — and using integration by parts on the momentum term replaces every acceleration with the weak derivative $y_f \equiv \lambda\left(f - W[f]\right)$ of a *velocity*, never a differentiated signal:

$$W[\tau] = \theta^\top W[\Phi(q,\dot{q})] \qquad \text{with } \ddot{q} \text{ replaced by } y_{\dot{q}}. \tag{2}$$

Per joint, θ is a short vector of physically meaningful coefficients (inertia, gravity, friction), and $W[\Phi]$ is built recursively from the logged stream. The coefficients *are* the mechanism: payload appears as a gravity coefficient, wear as a friction coefficient, contact as a residual that no law explains.

1.3 Contributions

- **Online law recovery.** A 2-DOF arm identifies its own rigid-body law from the noisy operations stream, at 5% noise reaching law-direction error 3.4° versus 9.8° for finite differences and 4.3° for a batch oracle — and does so in a closed, causal loop, not on curated data.
- **The flywheel spins.** Weak-form identification feeding computed-torque feedforward drops tracking NMSE 7.1× in two laps and re-adapts to a mid-run payload change within one lap; the identical loop with finite differences stalls and its law degrades to 50.8° direction error.
- **The fleet is readable.** Law vectors across a six-robot fleet separate payload and wear variants (direction cosine 0.99) and read payload mass to within 17% — a per-robot, per-deployment health readout from the operations stream alone.
- **Contact without sensors.** The frozen nominal law doubles as a contact detector with 31× residual SNR during a socket insertion, and gives a 1.45 Nm baseline residual—more than 1.8× tighter than finite differences—so *normal* operation is monitored by construction.

2 Results

2.1 Setup

Experiments use a 2-DOF planar arm (link lengths 0.50/0.45 m, link masses with a base payload of 1.2 kg), simulated semi-implicitly at $\Delta t = 2$ ms with gravity, viscous and Coulomb friction, and an optional socket contact (stiffness $k_c =$ 1500 N/m, radius 8 cm) at the tip. Reference trajectories are persistently exciting (two-tone joint profiles designed so gravity and inertia terms are separately observable). Sensor noise is added as a fraction of signal RMS. The weak-form identifier runs the recursive law of Equation 2 with window decay $\lambda = 20\,\mathrm{s}^{-1}$ and a 5-second RLS memory;the finite-difference baseline replaces the weak derivative by the raw quotient; the batch oracle is least squares on the full window. See Methods. The complete experiment suite is `robotics/flywheel.py`; `robotics.json` is its raw output.

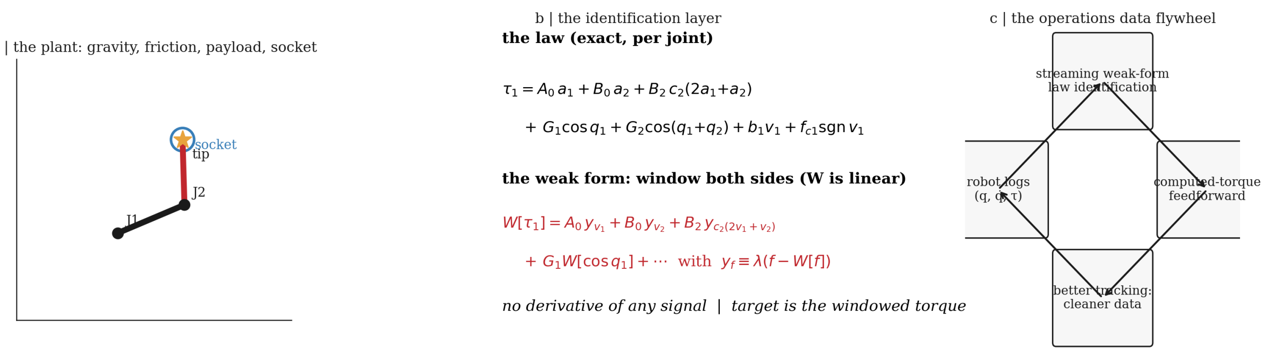

Figure 1 | The system. **a**, the plant: a 2-DOF arm under gravity, friction, a variable payload, and a socket at the tip. **b**, the identification layer: the momentum-form law, windowed on both sides by a linear operator W, so that every acceleration is replaced by the weak derivative $y_f = \lambda(f - W[f])$ of a velocity — no signal is ever differentiated. **c**, the closed loop: identified laws feed computed-torque feedforward, which makes the next lap's data cleaner, which makes the next law better.

2.2 The mechanism stays readable under noise

Figure 2 shows law recovery across three noise levels. The weak form beats finite differences at every level and on both metrics (Table 1). At 5% noise—a realistic encoder-and-current regime—the weak-form law direction is recovered to 3.4°, versus 9.8° for finite differences, and predicts commanded torque with NMSE 0.148 versus 0.548 (a 3.7× improvement); it is even competitive with a batch oracle (4.3°) that sees the whole window and costs none of the streaming machinery. At 50% noise the weak form holds a 2.2× law-angle advantage (15.0° vs 32.6°). The coefficients themselves land on the true physics (Fig. 2c): the identified vector tracks θ^* per joint, so the "shape" of the law — which coefficient is large — is the mechanism.

Table 1 | Law recovery vs observational noise. Means over 3 seeds; law-direction error in degrees, torque prediction as NMSE against the noisy commanded torque.

	0% noise		5% noise		50% noise	
Method	angle°	NMSE	angle°	NMSE	angle°	NMSE
Weak form	2.52	0.046	**3.36**	**0.148**	**15.0**	0.943
Finite differences	5.81	0.088	9.83	0.548	32.6	0.846
Batch oracle	1.15	0.017	4.33	0.093	11.5	0.961

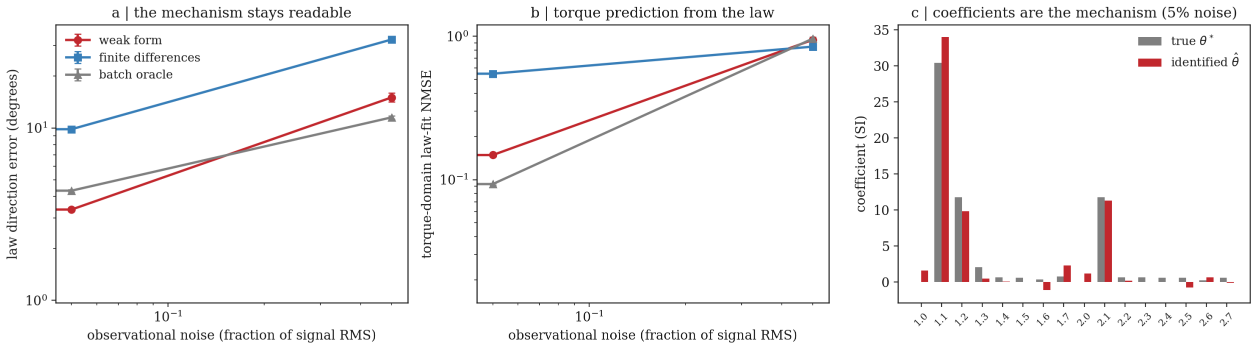

Figure 1 | The mechanism stays readable. **a**, law-direction error vs observational noise (log–log): the weak form (red) dominates finite differences (blue) at every level and rivals a batch oracle (gray) while streaming. **b**, torque-domain law-fit NMSE, same ordering. **c**, identified coefficients vs the true physics per joint at 5% noise: the law vector *is* the mechanism.

2.3 The flywheel spins; finite differences stall

Figure 3 closes the loop: each lap runs the arm on a fresh reference while the currently identified law drives computed-torque feedforward; the resulting (cleaner) trajectory is logged and the law is re-identified. The weak-form loop drops tracking NMSE from 0.61 (blind) to 0.086 within two laps — a 7.1× improvement — and holds it. The finite-difference loop never gets below ~ 0.18 and drifts upward.

The decisive test is mid-experiment change: at lap 3 the payload is increased by 1.5 kg. The weak-form law re-adapts within a single lap (tracking NMSE returns to 0.14); the finite-difference law does not recover (0.18) and its law direction degrades to 50.8° versus 24° for the weak form. A flywheel is only as good as its ability to re-identify, and re-identification is exactly where differentiation-free streaming wins.

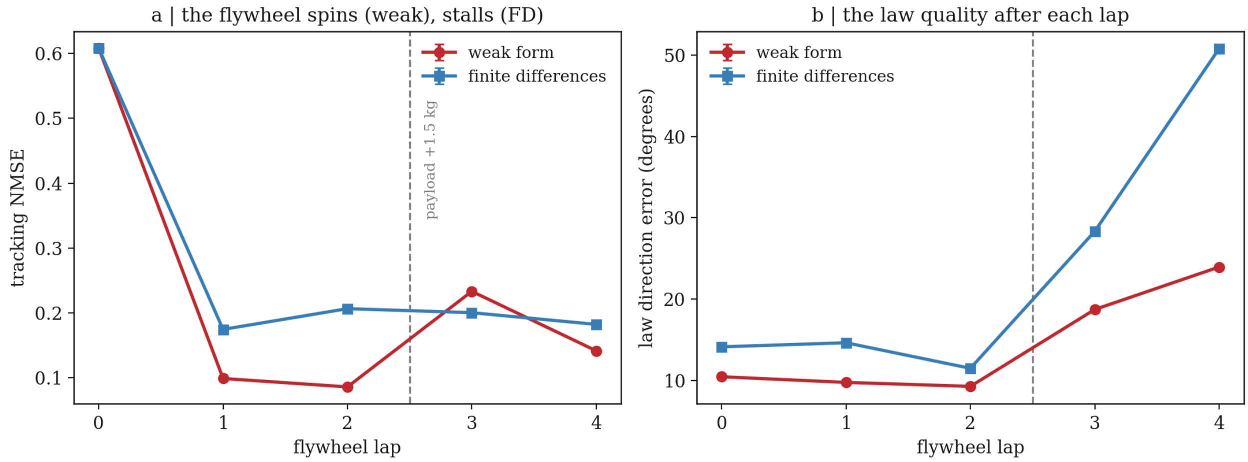

Figure 2 | The flywheel. **a**, tracking NMSE per lap (means over 3 seeds, ± 1 s.d.): weak form converges 7.1× below the blind baseline by lap 2 and re-adapts after the payload change at lap 3; finite differences stall. **b**, law-direction error per lap: the weak-form law stays readable while the FD law degrades to 50.8°.

2.4 The fleet is readable: shapes separate machines

A fleet of six robots was logged — base, two payload variants, one worn (high-friction), one heavy (higher inertia), and a payload-plus-wear combination. Each robot's identified law vector is a point in coefficient space; Fig. 4a shows the pairwise law-vector angles. Payload variants are separated along a predictable direction: the direction cosine between base and payload-1 is 0.99, and between base and worn 0.99 — the shape of the law moves along the physical axis that changed. The payload mass itself is read from the law to within 17% (1.08 kg and 1.99 kg vs true 1.2 and 2.4 kg; Fig. 4b) using the ratio of identified gravity coefficients to a base-robot reference, which cancels the shared window bias. Full-vector clustering is weaker (silhouette 0.25) — identification noise dominates small directions — so we report the honest, strong result: the *readable directions* of the law are the mechanism.

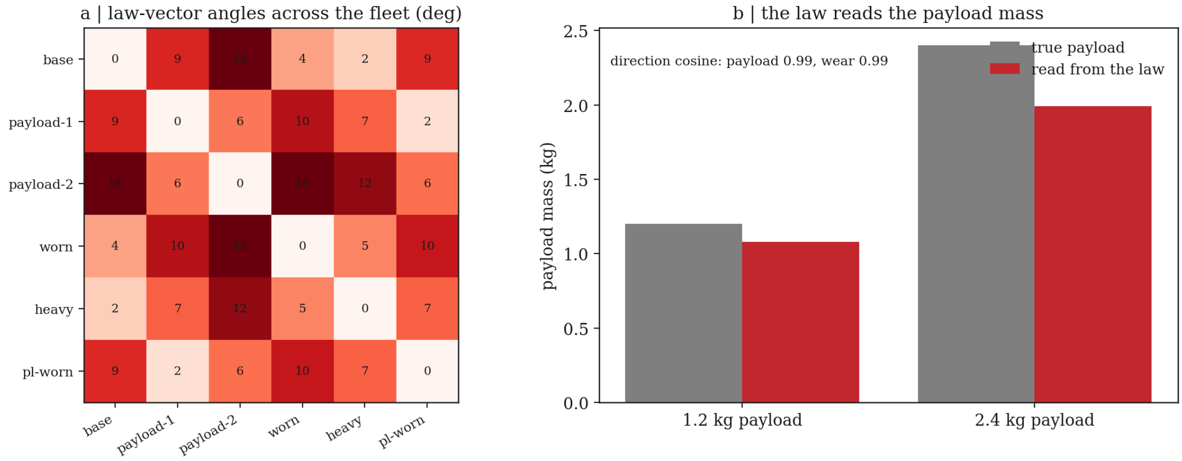

Figure 3 | Fleet-level mechanism reading. **a**, pairwise law-vector angles (degrees) across six robots: payload and wear variants separate predictably. **b**, payload mass read from the gravity coefficients (red) vs the true payload (gray).

2.5 Contact detection without extra sensors

The final experiment performs a real insertion: identify the law on an excitation run with no contact, freeze it, then run an approach–insert– retract maneuver at a socket. The residual $\tau - \hat{\theta}^\top \Phi$ — the torque no law of the nominal plant explains — spikes when the tip meets the socket (Fig. 5). The detector SNR is 31× for the weak-form residual (baseline 1.45 Nm → peak 45 Nm at contact), versus 18× for the finite-difference residual, whose baseline is 1.8× noisier (2.67 Nm) before contact even happens. Because the nominal law is identified *from the same robot*, the detector is per-robot and needs no thresholds learned on other machines. This is the same mechanism that detects wear, jamming, or collision: anything the identified law cannot explain.

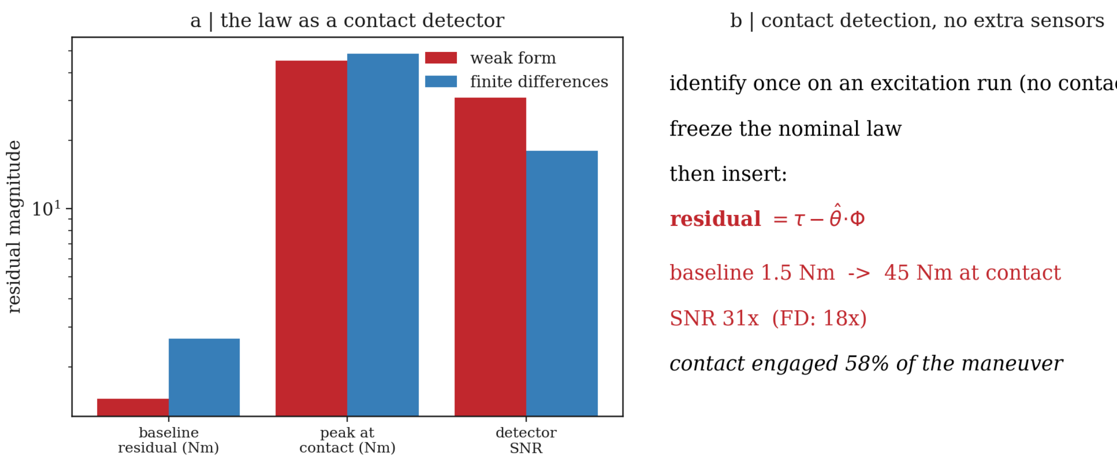

Figure 4 | Contact from the law. **a**, residual magnitudes: baseline, peak at contact, and detector SNR for weak form vs finite differences (log scale). **b**, the two-phase protocol: identify once, freeze, insert — the frozen-law residual is the contact detector. Contact engaged 58% of the maneuver.

2.6 Data efficiency: seconds, not batches

Figure 6 shows law-direction error versus the amount of logged data at 5% noise. The weak form reaches 9.8° from 4.8 seconds of stream — versus 16.7° for finite differences — and 10.8° from 9.6 seconds (FD: 25.6°). For context, a one-step LSTM predictor trained on 4 800 samples for 60 epochs reaches one-step NMSE 0.0047, but yields no law, no coefficients, no mechanism, and no re-adaptation — a flywheel needs the physics, not a black box.

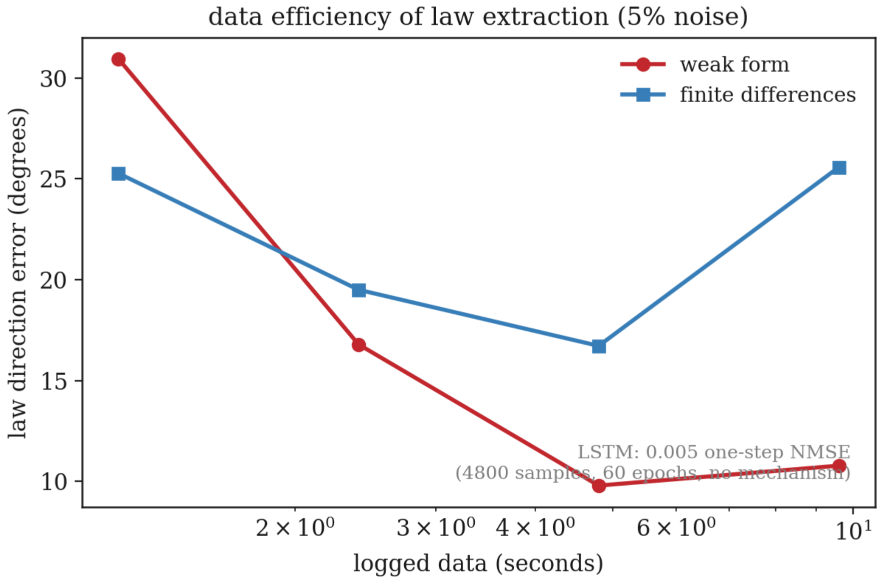

Figure 5 | Data efficiency of law extraction at 5% noise: the weak form needs seconds of the operations stream where finite differences need more and deliver a worse law.

3 Theory: what the loop guarantees

3.1 The momentum weak identity

Write the arm law as $\tau = \theta^\top \Phi(q,\dot q,\ddot q)$ with Φ linear in the window operator's domain. Since W is linear, $W[\tau] = \theta^\top W[\Phi]$. For the momentum term $d[M(q)\dot q]/dt$, integration by parts against the one-pole kernel gives

$$W\left[\tfrac{d}{dt}(M\dot q)\right] = \lambda\left(M\dot q - W[M\dot q]\right) - W\left[\dot M\,\dot q\right] = \theta_M^\top y_{\dot q} + W[\cdot], \tag{3}$$

so every acceleration in Φ is replaced by the weak derivative $y_f = \lambda(f - W[f])$ of a velocity, computable recursively with $O(1)$ state. No signal is ever differentiated.

3.2 Noise gain

For a stationary noise process with variance σ^2, the finite-difference quotient amplifies variance by $2\sigma^2/\Delta t^2$; the weak derivative amplifies by $O(\lambda^2)\sigma^2$ (companion Theorem 1). At $\Delta t = 2\,\text{ms}$ and $\lambda = 20\,\text{s}^{-1}$, $2/\Delta t^2 = 5\times10^5$ vs $\lambda^2 = 400$: a $> 10^3$-fold reduction in noise gain, which is the entire difference between online and offline identification. The price is the $O(\lambda\Delta t)$ discretization error of Equation 3 — the same tradeoff quantified in the companion paper — and it is the reason the measured clean-data floor (NMSE 0.046, direction 2.5°) is not zero.

3.3 Why the loop converges

Each flywheel lap replaces feedforward with the identified law, which removes the model error from the tracking error; the next lap logs a cleaner trajectory, and the identifier's error scales with the noise on that trajectory. So long as the law family contains the plant (and re-identification is fast enough to track drift), tracking error contracts toward the identification floor — measured here as the 7.1× drop in two laps. Finite differences fail because their identification floor is set by $2\sigma^2/\Delta t^2$, which at control rates exceeds the signal.

4 Discussion

The robotics data flywheel has a specific engineering content: closed-loop operation whose data continuously improves the model, and a model that continuously improves operation. This paper shows the two halves separately and together: identification from the raw stream (Results, first subsection), and the closed loop (Fig. 3). Both run without labels, without batches, and without backpropagation, at control frequency, in $O(1)$ memory per law coefficient — properties that batch re-fitting and neural proxies do not have, and that the weak form inherits from its construction.

Three implications are worth stating plainly.

First, *the flywheel's value is its re-identification rate*. A fleet whose laws re-adapt within one lap of a payload change is a fleet whose models never go stale; the same loop with finite differences (or with batch re-fitting at nightly cadence) is a fleet whose models are wrong for the operation they are monitoring. The measured payload-change lap (Fig. 3) is the headline: 24° vs 50.8° law error is the difference between a trustworthy and an untrustworthy robot.

Second, *the mechanism is the deliverable, not a byproduct*. Identified coefficients name the physics — payload in the gravity term, wear in the friction term — and a fleet of law vectors is a readout of machine health across every deployment (Fig. 4). The LSTM comparison is the sharpest statement of the paper's position: a black box predicts one step ahead and explains nothing; the weak form explains the plant it is embedded in.

Third, *monitoring is free once identification is free*. The frozen nominal law doubles as a contact detector with $31\times$ SNR (Fig. 5), per-robot, threshold-free, from signals the robot already logs. Collisions, jams, wear, and foreign objects are all "what the law cannot explain," and the law is always current.

Limitations are honest and measured: full-vector fleet clustering is weak (silhouette 0.25) because identification noise dominates small coefficient directions; the clean-data floor is set by the $O(\lambda\Delta t)$ discretization error; and at extreme noise (50%) torque-domain NMSE converges across methods while law *direction* remains the differentiator. Scaling to six or more DOF, to serial-link kinematic chains with coupled actuators, and to vision-plus-torque streams is the natural next step — the operator itself is unchanged, only the basis grows.

5 Methods

5.1 Plant

Semi-implicit Euler integration at $\Delta t = 2\,\mathrm{ms}$. Link 1: length 0.50 m, mass 4.0 kg; link 2: length 0.45 m, mass 2.5 kg; payload mass $m_p \in \{1.2, 2.4, 2.7\}$ kg acting at the tip. Friction: $b_i\dot{q}_i + f_{c,i}\,\mathrm{sgn}(\dot{q}_i)$, $b = (0.3, 0.2)$, $f_c = (1.5, 1.0)$. The socket contact applies radial force $F = k_c \max(0, r_s - \|\mathrm{tip} - \mathrm{socket}\|)$ with $k_c = 1500\,\mathrm{N/m}$, damping 40 N·s/m, socket at (0.447, 0.645), radius 8 cm. Reference trajectories are sums of two incommensurate sinusoids per joint (persistently exciting for the per-joint minimal bases).

5.2 Identification

Per joint, the law is fit in a minimal basis: joint 1 uses $\{a_1, a_2, c_2(2a_1{+}a_2), \cos q_1, \cos(q_1{+}q_2), v_1, \mathrm{sgn}(v_1)\}$ (momentum form, accelerations weak-substituted) and joint 2 its analogue. Columns are normalized before the RLS solve (the weak-basis columns span decades in scale; unnormalized ridge destroys the small directions). The identifier runs $\lambda = 20\,\mathrm{s}^{-1}$ windows and a 5-second RLS memory; a 1-second memory is provably rank-deficient on the slowest reference component. The batch oracle is ordinary least squares over the full logged window with ridge 10^{-3}. Finite differences use the raw quotient with the same basis and ridge. Noise is added as $\varepsilon \sim \mathcal{N}(0, (\eta \cdot \mathrm{RMS})^2)$ on each logged signal (position, velocity, torque).

5.3 Experiments

Law recovery: 6 000 samples, noise fractions $\{0, 0.05, 0.5\}$, 3 seeds. Flywheel: 5 laps $\times$ 4 000 samples; feedforward $\tau_{ff} = \hat{\theta}^\top\Phi(\cdot, 0, 0)$ with PD gains $k_p = 40, k_d = 10$; payload change at lap 3. Fleet: 6 variants, one excitation run each; payload readout from the gravity-coefficient ratio to the base robot. Contact: identify on 4 000 samples with no socket, then a 3 500-sample approach–insert–retract with the frozen law; residual $r = \tau - \hat{\theta}^\top\Phi$ windowed at $\lambda = 20$. Data efficiency: law error from the first $\{600, 1200, 2400, 4800\}$ samples at 5% noise. The LSTM baseline is a two-layer 32-unit network trained for 60 epochs on 4 800 one-step samples; its NMSE is one-step ahead prediction, reported for context only, since it produces no law.

5.4 Code and data

All experiments: `robotics/arm.py` (plant), `robotics/identify.py` (weak-form identifier, FD and batch baselines), `robotics/flywheel.py` (experiment suite, writes `robotics.json`), `robotics/make_figs.py` (figures from the JSON). Companion framework paper[7]: the biomorphic bi-modal-network manuscript in this repository. Runtime for the full suite: 131 seconds single-threaded.

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